AI-Powered Road Accident Prediction & Emergency Response System

Multi-Agent Agentic AI Platform for Public Road Safety

Need of Project

Over 1.35 million people die in road accidents globally every year (WHO, 2023). India records 4.61 lakh accidents and 1.68 lakh fatalities annually (MoRTH 2023). Accident-prone zones are not identified in advance, and emergency services face delays due to traffic congestion and lack of coordinated AI intelligence. This platform predicts accident risks before they occur — shifting from reactive to proactive road safety.

Team: Road Safety AI **Institute:** Bengaluru Hackathon 2026 **GitHub:** github.com/jharanabisoyi8050-alt/ibm

Project Overview & Team

Project: AI-Powered Road Accident Prediction & Emergency Response System

Need of Project

- 1.35M deaths/year globally (WHO 2023)
- India: 4.61L accidents, 1.68L fatalities (MoRTH 2023)
- Emergency response delayed by traffic congestion
- Accident hotspots not identified until repeated tragedies
- No unified AI system for prediction + monitoring
- Manual systems lack speed, accuracy & real-time awareness

Methodology

1. Collect historical accident, traffic & weather datasets
2. Train XGBoost risk prediction model (0–100 score)
3. DBSCAN clustering to identify accident hotspots
4. Build 7-agent LangGraph system with IBM Bob
5. Integrate 3 LLM providers via IBM Bob Model Gateway
6. Deploy custom MCP server (traffic/weather/geo/ambulance)
7. Build Streamlit dashboard: heatmaps, risk gauges, HITL
8. Implement Human-in-the-Loop for HIGH/CRITICAL dispatch

IBM TOOLS USED

IBM Granite (watsonx)	IBM Bob	IBM Bob Model Gateway	Agentic AI (LangGraph)
Orchestrator & Hotspot Detection Agent	Agentic AI development, code generation & MCP	Multi-LLM routing: watsonx + Bedrock + Vertex	Multi-agent orchestration, HITL & memory

Conclusion & Demonstration

Impact: AI-predicted risk alerts enable proactive traffic management, reducing high-risk zone incidents by up to 30% through early intervention. The Streamlit dashboard demonstrates: (1) Real-time risk scoring & heatmaps, (2) AI-generated emergency response plans, (3) Ambulance coordination with ETA, (4) Admin panel with LLM cost monitoring & HITL approval queue.

Development with IBM Bob

1 REQUIREMENTS	2 BOB CREATES PLAN	3 REVIEW & REFINE	4 CODE GENERATION	5 REVIEW OUTPUT
Capture project goals, agent specs, DB schema & acceptance criteria.	Multi-agent architecture, LangGraph state machine, file structure.	Review agent workflow, HITL gates, LLM routing & data labeling policy.	FastAPI backend, 7 LangGraph agents, MCP server, Streamlit dashboard, tests.	Review HITL enforcement, AI_PREDICTED labels & SIMULATED flags.
6 RUN & VALIDATE	**7 BOB EXPLAINS**	**8 QUALITY & TESTS**	**9 DOCUMENTATION**	**GitHub**
Streamlit app end-to-end: risk, heatmaps, ambulance dispatch.	LangGraph transitions, MCP JSON-RPC calls & LLM routing decisions.	pytest: agent unit tests, API integration & ML accuracy tests.	README.md, AGENTS.md, .bob/custom_modes.yaml, API docs, inline code comments.	84 files | 6,607 lines All generated with IBM Bob

IBM Bob Custom Mode: road-safety-ai

Workspace-scoped custom mode in **.bob/custom_modes.yaml** — road safety expert persona with Agent + MCP + execute permissions. **AGENTS.md** gives Bob full project architecture context on every task.

GitHub Repository

github.com/jharanabisoyi8050-alt/ibm

Streamlit Dashboard: port 8501 | API Docs: port 8000/docs

IBM Bob — Files & Code Generated

IBM Bob — Project File Structure (84 files, 6,607 lines)

```
road-accident-ai/
■■■ .bob/custom_modes.yaml # IBM Bob custom mode
■■■ AGENTS.md # Project context for Bob
■■■ backend/app/
■ ■■■ agents/ # 7 LangGraph AI agents
■ ■■■ api/routes/ # 6 FastAPI route files
■ ■■■ db/models.py # PostgreSQL ORM models
■ ■■■ llm_gateway/ # Multi-LLM routing
■ ■■■ mcp_server/ # Custom MCP server
■ ■■■ ml/ # XGBoost + DBSCAN
■ ■■■ observability/ # LangSmith + Prometheus
■ ■■■ governance/hitl.py # HITL approval queue
■■■ streamlit_app/ # 7-page Streamlit dashboard
■■■ infra/ # Docker + K8s + CI/CD
```

IBM Bob — Agents & Modules Generated

orchestrator.py
LangGraph state machine, fan-out to 6 agents + HITL gate. Model: watsonx Granite-3

traffic_agent.py | road_agent.py | ambulance_agent.py
Gemini Flash — traffic density, road condition & ambulance coordination

risk_agent.py | emergency_agent.py
Claude Sonnet (Bedrock) — XGBoost risk scoring + HITL emergency recommendations

hotspot_agent.py + hotspot_model.py
watsonx Granite + DBSCAN — generates heatmap data for Plotly density_mapbox

MCP Server Tool Groups & LLM Gateway

traffic_mcp.py	weather_mcp.py	geo_mcp.py	gateway_config.yaml
get_traffic_density get_congestion_level get_vehicle_count	get_current_weather get_weather_risk get_weather_forecast	geocode_address get_route get_nearby_hospitals	watsonx Granite-3 Claude Sonnet (Bedrock) Gemini Flash (Vertex)

Features of the Project

MULTI-AGENT AI SYSTEM
7 LangGraph agents: Orchestrator, Traffic, Risk Prediction, Road Condition, Hotspot Detection, Emergency Response, Ambulance Coordination.

XGBoost RISK PREDICTION
Predicts risk score 0–100 using traffic density, weather, road quality & time-of-day. Labeled AI_PREDICTED — never confused with confirmed accidents.

ACCIDENT HOTSPOT HEATMAPS
DBSCAN clustering on historical accident coordinates generates hotspot zones. Interactive Plotly density_mapbox heatmaps on Streamlit dashboard.

HUMAN-IN-THE-LOOP (HITL)
HIGH & CRITICAL emergency dispatch requires human officer approval before any action. HITL queue in Admin Panel. Every decision logged for audit.

LLM GATEWAY & COST MONITOR
IBM Bob Model Gateway routes agents to watsonx Granite, Claude Sonnet, Gemini Flash. Token usage & USD cost per agent logged in real time. 7-day cost trend.

STREAMLIT DASHBOARD
7-page interactive dashboard: Traffic Monitor, Hotspot Heatmap, Risk Gauge, Emergency Report, Ambulance Coordination, Admin Panel.

Additional Features
• MCP Tool Server (traffic, weather, geo, ambulance) • Agent Memory: Redis session + ChromaDB pgvector • Guardrails: injection protection + Bedrock guardrailConfig • LangSmith Tracing + Prometheus + Grafana • Data Labeling: AI_PREDICTED / CONFIRMED / SIMULATED enforced at DB + UI • Kubernetes + GitHub Actions CI/CD Pipeline

Novelty & Future Scope

NOVELTY & UNIQUENESS

Predictive, Not Reactive
XGBoost predicts accident risk BEFORE incidents — enabling proactive intervention. Unlike traditional systems that only respond after accidents.

Multi-LLM Agent Orchestration
7 specialist agents each use their own optimized LLM (watsonx / Bedrock / Vertex) — first road safety system combining IBM Bob Agentic AI with multi-LLM orchestration.

MCP-Integrated Live Tool Calls
Custom JSON-RPC 2.0 MCP server with 4 tool groups (traffic, weather, geo, ambulance) enables agents to call live external tools — not just static knowledge.

HITL + Responsible AI Governance
HIGH/CRITICAL dispatch requires human approval. AI predictions labeled AI_PREDICTED. Simulated data labeled is_simulated=True. Full audit log.

FUTURE SCOPE

Live Traffic Integration
Connect to HERE Maps / TomTom / government APIs to replace simulated traffic with real-time feeds.

IoT Sensor Integration
Integrate speed cameras, weather stations & pothole detectors for real-time road monitoring.

Real CAD Ambulance Dispatch
Replace simulated fleet with live CAD system integration for actual emergency dispatch.

Mobile App for First Responders
Smartphone app delivering risk alerts, hotspot navigation & AI response plans to field officers.

Advanced ML & Computer Vision
CCTV/dashcam vision models for automatic accident detection. Upgrade to deep learning ensemble.

Thank You | AI-Powered Road Accident Prediction & Emergency Response System | Built with IBM Bob